

Experiment 4: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1977	11.497	13.538	21.933	13.1%	26.4%	31.7%	8.455	17.837	5.414	11.767	0.408	0.983	1.303
Proposed	1930	2.764	1.067	8.646	46.3%	72.3%	80.9%	16.307	30.891	7.928	17.172	0.529	1.312	1.474
PPO-GRU	1933	2.948	1.186	9.417	43.0%	69.3%	80.2%	14.722	29.919	7.437	16.134	0.486	1.239	1.434
MPC	941	4.891	2.626	14.349	2.1%	20.4%	71.1%	2.496	3.964	1.512	2.728	0.143	0.150	0.181

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1977	98.8%	23	1.1%	1.1%	23	100.0%	0	0
Proposed	2000	1930	96.5%	70	3.5%	3.5%	70	100.0%	0	0
PPO-GRU	2000	1933	96.7%	67	3.4%	3.0%	59	88.1%	8	0
MPC	2000	941	47.0%	1059	53.0%	53.0%	1059	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	1059	1041	98.3%	752	72.2%	11.956
Proposed	1059	1000	94.4%	232	23.2%	3.104
PPO-GRU	1059	1000	94.4%	231	23.1%	3.343